

Short Report: Decision Tree vs Naïve Bayes (Iris)

➔ Dataset

- Used uploaded CSV: iris-2.csv.
- Classes: ['setosa', 'versicolor', 'virginica']

➔ Model Evaluation (10-fold CV on full data)

- Naïve Bayes: Acc=0.953, Prec=0.953, Rec=0.953, F1=0.953
- Decision Tree (entropy): Acc=0.933, Prec=0.934, Rec=0.933, F1=0.933

➔ Holdout Split (70/30 stratified)

- Naïve Bayes: Acc=0.911, Prec=0.916, Rec=0.911, F1=0.911, Macro-AUC=0.989
- Decision Tree (entropy): Acc=0.889, Prec=0.890, Rec=0.889, F1=0.889, Macro-AUC=0.917

➔ Interpretability

- Decision Tree provides a visual hierarchy of splits; we also exported its path-based IF-THEN rules (see rules_from_tree.txt / rules_screenshot.png).
- Naïve Bayes offers simple probabilistic reasoning with class-conditional likelihoods and the strong (often violated) feature-independence assumption.

➔ When to Use Which

- Rule/Tree-based: when interpretability and human-readable logic is key; can capture non-linear, interaction-heavy patterns but may overfit without pruning.
- Naïve Bayes: when features are roughly independent or high-dimensional/sparse (e.g., text); fast, robust with small data, but decision boundaries can be simplistic.

➔ Optional Bootstrap (NB on resampled data)

Bootstrapped accuracy mean=0.957 (std=0.006) over B=200 resamples.

➔ Top Splitting Features (entropy tree)

petal_length: 0.623

petal_width: 0.337

sepal_length: 0.028

sepal_width: 0.012